

GYUYEON LIM

B.S. Candidate, Computer Engineering
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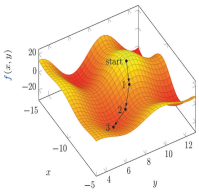
education

2023–2027 **B.S. in Computer Engineering** (Advanced Major Track), Gachon University, Seongnam-si (expected).

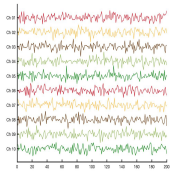
- GPA: **4.41 / 4.5** (Percentile **99.10%**; 121 of 120 credits)
- Research guidance from Prof. Myung-Kyu Yi, Institute of Biomedical Engineering, Hanyang University (2024.06–2025.12)
- Graduation thesis advised by Prof. Yourim Yoon, Gachon University

2020–2023 **Mathematics & Science Intensive Program**, Hyosung High School, Seongnam-si.

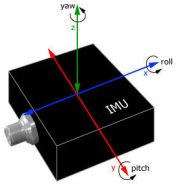
research interests



Machine Learning. Learning compact, structured, and robust embeddings from high-dimensional data. Focuses on robust representations that remain valid in unseen environments by disentangling shared, invariant information from observation-dependent variation, together with contrastive and self-supervised methods, data embedding, and latent space geometry.



Temporal or Spatiotemporal Modeling. Temporal and spatiotemporal modeling for forecasting and classification of multivariate sensor streams. Covers sequence models, graph-based dependencies, and large-scale time-series architectures.



AI for Science. Applying deep learning to physical and biological signals — IMU, EEG, and multivariate sensor data. Integrates physics priors, domain constraints, and signal structure into model design.

publications (conference submissions)

Preprinted Junha Kim*, **Gyuyeon Lim***, and Jay-Yoon Lee. "STREAM: Dynamic Channel Graphs with Autoregressive State Propagation for Multivariate Forecasting." *submitted*.

Preprinted Taeyang Lee*, **Gyuyeon Lim**, Sebin Lee, Kevin Valencia, Jay-Yoon Lee, Xihaier Luo, Shinjae Yoo, Jiook Cha, and David Keetae Park.

"NeuroX-EEG: Scalp Field Inpainting for Zero-Shot EEG Super-Resolution." (2nd author) *submitted*.

Preprinted — Yidu Song*, **Gyuyeon Lim***, and Ahyoung Choi. "ENURI: Evidence Negotiation Using Reasoning over Images." *submitted*.

publications (journal articles)

2026.09 — **Gyuyeon Lim*** and Yourim Yoon. "Global-Local Dual Prototype Network for Interpretable Time Series Classification." *Asia-Pacific Journal of Convergent Research Interchange*, Vol. 12, No. 9, pp. 293–305 (2026). DOI.

2026.07 — **Gyuyeon Lim*** and Myung-Kyu Yi. "LPSN: A Learnable Parametric Spectral Network for Wearable Human Activity Recognition." *IEEE Sensors Journal*, Early Access (IF 4.5). DOI.

2026.04 — **Gyuyeon Lim*** and Myung-Kyu Yi. "Rotation-Robust Self-Supervised Learning for IMU-Based HAR with Multi-Scale Convolutional Backbones." *IEEE Transactions on Consumer Electronics* (IF 10.9). DOI • Code (InceptionMK).

2026.03 — JunYoung Park*, **Gyuyeon Lim**, and Myung-Kyu Yi. "Dynamics-Aware Action State Flow for Wearable Human Activity Recognition." *IEEE Sensors Journal* (IF 4.5). (2nd author) DOI.

2026.03 — **Gyuyeon Lim*** and Myung-Kyu Yi. "Robust Sensor-Based Activity Recognition via Physics-Aware Disentangled Representation Learning and Feature-wise Linear Modulation Conditioning." *IEEE Sensors Journal* (IF 4.5). DOI • Code (PADR).

2026.02 — **Gyuyeon Lim*** and Myung-Kyu Yi. "CAP-SSM: Context-Aware Prototypes and Selective State-Space Modeling for Efficient Wearable Human Activity Recognition." *IEEE Sensors Journal* (IF 4.5). DOI • Code (CAP-SSM).

2026.01 — **Gyuyeon Lim*** and Myung-Kyu Yi. "Prototype-Guided Physics-Modulated Perceiver for Human Activity Recognition." *IEEE Internet of Things Journal* (IF 8.9). DOI • Code (P2-Perceiver).

2025.09 — **Gyuyeon Lim*** and Myung-Kyu Yi. "Contrastive Learning for Zero-Shot Human Activity Recognition Using Labeled Simple Actions on Wearable Devices." *IEEE Sensors Journal* (IF 4.5). DOI • Code (ZSL-HAR).

research experience

2025.12–Present — **Research Intern**, Seoul National University Graduate School of Data Science, SKI-ML (Prof. Jay-Yoon Lee), Seoul.

- Structure and Knowledge Injection into Machine Learning (SKI-ML lab)
- Spatio-temporal multivariate modeling for large-scale time-series with graph dependencies (STREAM)
- Co-authored NeuroX-EEG, in collaboration with the Connectome Lab (Prof. Jiook Cha)

2025.02–2025.06 — **Undergraduate Research Intern**, Seoul National University Hospital, BMI Lab (Prof. Kwangsoo Kim), Yeongeon, Seoul.

- Biomedical Informatics Lab (SNUH BMI LAB)

- Holter Foundation Model and MLOps Development Team

academic service

2026–Present **Reviewer**
- *IEEE Internet of Things Journal*
- *Informatics and Health*

teaching experience

2025.09–Present **Math-based University Essay Instructor**, Nonsoolmate, Seoul (Remote).
- Grading and coaching students on math-based university admissions essays

2026.01–2026.08 **High School Math Instructor**, Myungmun Seoil Academy Premium, Seongnam-si.
- CSAT (Suneung) and school exam preparation, STEM track focus

2023.06–2026.06 **Lead Math Instructor**, Metrix Mathematics Institute (HB Math&Eng), Seongnam-si.
- Primary instructor for middle and high school students (grades 2–3) and CSAT retakers
- Specializing in CSAT preparation for STEM track students

scholarships

2025.12 **SCI Publication Scholarship**, Gachon University — the Highest Honor.

Every semester **Academic Excellence Scholarship**, Gachon University.

From 3rd year **National Science & Technology Scholarship**, Ministry of Science and ICT (MSIT).

2024.01 **Science & Engineering Excellence Scholarship**, Anyang-Gwacheon Chamber of Commerce and Industry.

honors & awards

2026.08 SNU AI Challenge 2026 — **Final Rank 21st / 129 Teams** (Team Gyuyeon Lim, solo participant). "Temporal Event Matching via Permutation-scored Ordering Hard-Negative Listwise Ranking for Chronological Frame Reconstruction with Vision-Language Models." Report.

2026.08 **Research Division 2nd Place**, 1st Gachon University Computer Engineering–Artificial Intelligence Academic Festival, Gachon University. ENURI.

2026.01 P-Career Catch 2 — **Outstanding Completion**, Career Exploration Credit Program, Gachon University. GitHub.

2025.05 **Severance Challenge Award**, 2025 Yonsei Yongin Severance Hospital Healthcare Hackathon.

2025.06 **3rd Place**, AI/SW Day Oral Presentation, Gachon University — YOLOv5 Children Protection Area detection. GitHub.

2024.10 **ICPC Seoul Regional Finalist**, Team GPT — 168th of 376 teams, KINTEX, Goyang-si. Official.

2024.01 **Grand Prize**, Future Design Project, Gachon University (1st place).

2023.06 **Grand Prize**, Software Competition, Gachon University (1st place) — Smart Home with Physical Computing.

2018.11 **Minister of Oceans and Fisheries Award**, 2018 National Student Science and Invention Competition.

certifications

2026.09 *Language.* **TEPS** (Test of English Proficiency developed by Seoul National University) — 340 (Grade 2).

2025.05 *Technical.* **AICE Associate** (AI Certificate for Everyone), KT.